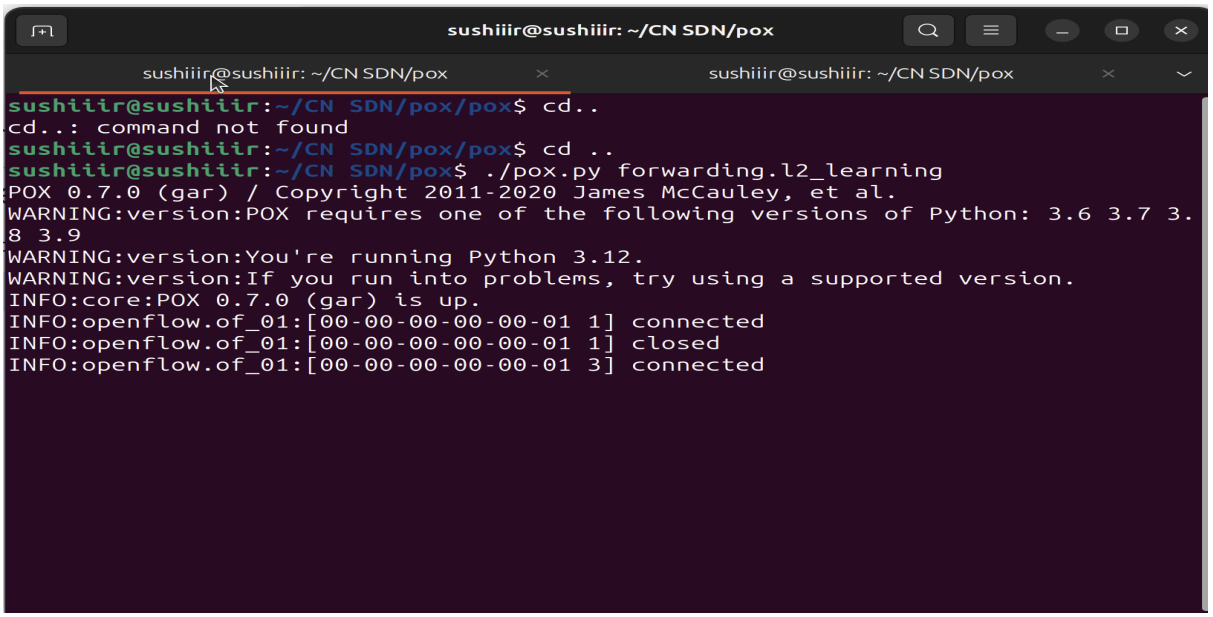
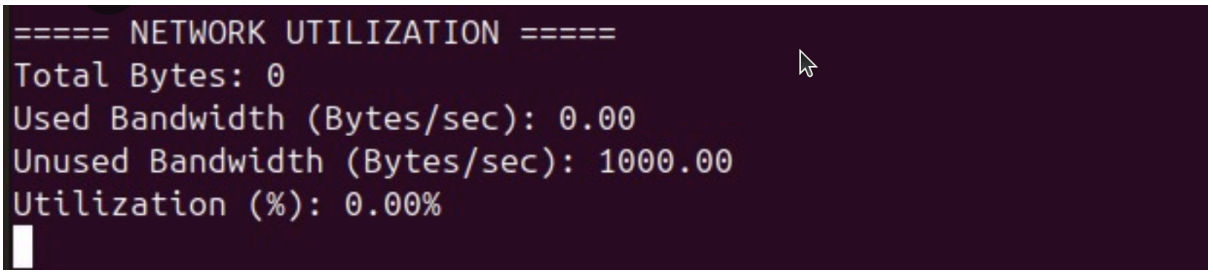
# SDN Network Utilization Monitor: Output Screenshots

## Project Overview

**Name:** Sushir C (PES1UG24AM297)



sushiiir@sushiiir: ~/CN SDN/pox

sushiiir@sushiiir: ~/CN SDN/pox | sushiiir@sushiiir: ~/CN SDN/pox

```
sushiiir@sushiiir:~/CN SDN/pox$ sudo mn -c
sudo mn --topo single,3 --controller remote,ip=127.0.0.1,port=6633
[sudo] password for sushiiir:
*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes
killall controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd
ovs-controllerovs-testcontroller udpbwtest mnexec ivs ryu-manager 2> /dev/null
killall -9 controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflo
wd ovs-controllerovs-testcontroller udpbwtest mnexec ivs ryu-manager 2> /dev/nul
l
pkill -9 -f "sudo mnexec"
*** Removing junk from /tmp
rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log
*** Removing old X11 tunnels
*** Removing excess kernel datapaths
ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'
***  Removing OVS datapaths
ovs-vsctl --timeout=1 list-br
ovs-vsctl --if-exists del-br s1
ovs-vsctl --timeout=1 list-br
*** Removing all links of the pattern foo-ethX
ip link show | egrep -o '([-_.[:alnum:]]+-eth[[:digit:]]+)'
ip link show
*** Killing stale mininet node processes
pkill -9 -f mininet:
```

sushiiir@sushiiir: ~/CN SDN/pox

sushiiir@sushiiir: ~/CN SDN/pox | sushiiir@sushiiir: ~/CN SDN/pox

```
ip link show
*** Killing stale mininet node processes
pkill -9 -f mininet:
*** Shutting down stale tunnels
pkill -9 -f Tunnel=Ethernet
pkill -9 -f .ssh/mn
rm -f ~/.ssh/mn/*
*** Cleanup complete.
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet>
```

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet>
```

```
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=5.08 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.16 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.239 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.101 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.321 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.091 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.276 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.138 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.218 ms
^C
--- 10.0.0.2 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8144ms
rtt min/avg/max/mdev = 0.091/0.847/5.076/1.526 ms
```

```
===== NETWORK UTILIZATION =====
Total Bytes: 442
Used Bandwidth (Bytes/sec): 88.39
Unused Bandwidth (Bytes/sec): 911.61
Utilization (%): 8.84%

===== NETWORK UTILIZATION =====
Total Bytes: 2318
Used Bandwidth (Bytes/sec): 463.46
Unused Bandwidth (Bytes/sec): 536.54
Utilization (%): 46.35%

===== NETWORK UTILIZATION =====
Total Bytes: 322
Used Bandwidth (Bytes/sec): 60.98
Unused Bandwidth (Bytes/sec): 939.02
Utilization (%): 6.10%

===== NETWORK UTILIZATION =====
Total Bytes: 210
Used Bandwidth (Bytes/sec): 40.48
Unused Bandwidth (Bytes/sec): 959.52
Utilization (%): 4.05%
```

```
mininet> h1 iperf -s &
mininet> h2 iperf -c h1
------------------------------------------------------------
Client connecting to 10.0.0.1, TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.2 port 48326 connected with 10.0.0.1 port 5001 (icwnd/mss/irt
t=14/1448/6877)
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.0130 sec  96.1 GBytes  82.4 Gbits/sec
mininet>
```